

with the AP for every guess. An alternate (and faster) method of validating a guess is to use a 4-way handshake. All that is needed is to capture the packets transmitted when a valid client connects. If these packets (the 4-way handshake) are available, the passwords can be validated against them. There are a few different ways of guessing the password:-

i. Bruteforcing

The software will try all possible combinations for the passwords. Time taken to crack the password increases exponentially with time. Brute force attacks work best if the WPS pin is still enabled in the router.

Here's what Wikipedia says about WPS-

“Created by Cisco and introduced in 2006, the point of the protocol is to allow home users who know little of wireless security and may be intimidated by the available security options to set up Wi-Fi Protected Access, as well as making it easy to add new devices to an existing network without entering long passphrases. Wi-Fi Protected Setup allows the owner of Wi-Fi privileges to block other users from using their household Wi-Fi. The owner can also allow people to use Wi-Fi. This can be changed by pressing the WPS button on the home router.

A major security flaw was revealed in December 2011 that affects wireless routers with the WPS PIN feature, which most recent models have enabled by default. The flaw allows a remote attacker to recover the WPS PIN in a few hours with a brute-force attack and, with the WPS PIN, the network's WPA/WPA2 pre-shared key (PSK). Users have been urged to turn off the WPS PIN feature, although this may not be possible on some router models.”

It is an eight-digit verification number to access the network. Validation of keys takes time, almost one second per key. So theoretically it would take 108 seconds, which is close to three years, but there is an easy workaround. The 8th digit is a checksum digit. So we only have to check the first seven. The PIN goes in two halves for verification. The first half would take only 104 guesses and the second half would take a mere 103 guesses.

So a total guess needed would be 11,000, which would take close to three hours and much lesser on current-gen hardware.

ii. Wordlist / Dictionary attacks

There is a predefined list of guesses (a word list or dictionary of passwords). These word list files contain commonly used passwords, combinations of passwords, misspellings of words, a combination of words and numbers